05-20-00 Scheduled Maintenance Checks

This section describes the scheduled maintenance checks for the E4 Series.
The scheduled annual inspection consists of the 100 hours inspection program.

	Activities/Service Interval	100 h	300 h
	General		
0.	Send the following data to Austro Engine GmbH: – Serial Number of the Engine(s) – Serial Number of the Aircraft – Aircraft Registration Marks – Company	X	X
1.	Review the applicable Airworthiness Directives.	X	X
2.	Review the applicable Service Bulletins.	X	X
3.	Review the Life-Limited Parts list. Refer to section 05-10-00.	X	X
4.	Review the Component Maintenance Requirements.	X	X
5.	Visually examine the whole engine surface using a flashlight and a mirror. – Look especially for signs of wear and leakage at the oil sump gasket and the cylinder head gasket. – Examine for loose parts or foreign objects.	X	X
	ELECTRIC		
6.	Examine the glow plug control. Refer to chapter 80.	X	X
7.	Examine the alternator according to chapter 24.		X
8.	Visually examine the wiring harness for signs of wear or damage. – Make sure that the camshaft sensor connection and the injector connections are tight fitted. Examine the engine harness for signs of wear or chafing against any structure. Make sure that the engine harness does not rub against the aircraft structure (e.g. pipes) between the primary attachment point (main clamp) and X1/1, X1/2 connectors. Refer to section 71-50-00 para 2.A. Make sure that the cables or connectors are not damaged or too much force has been applied (especially cowling in the area of high-pressure pump). Refer to section 71-50-00 para 2.A. Make sure that the BPA connector and the cable branch dedicated to the BPA are tightened (refer to Fig. 71-67).	X	X
9.	Visually examine the wiring harness that is not covered by the injector cover for signs of wear or damage. Make sure that the sensor connections are fitted tight. CAUTION: To prevent any damage on cables and sensors, do not put too much force!	X	X

	Activities/Service Interval	100 h	300 h
10.	Read out EECU-data using the AE 300 Wizard. For these downloading activities the engine must not be started (0 propeller RPM). Send the following data to Austro Engine GmbH (AE300data@austroengine.net): – Engine log – Event recorder – Engine data log (at least the last 3h of engine operation) – Perform a LiveView recording of about 20 seconds with the LiveView-Configuration file: „EMM_Engine_Check_LiveView_V1" or subsequent. (this file can be downloaded from the Diamond Partner Portal) Note: Once the files have been sent, the system will reply with a confirmation message. Include NOCONFIRM in the email subject to avoid this reply. Include REPLYEXCEL in your e-mail subject to receive Engine Logs and Event Records converted into EXCEL format. Or include REPLYEXCELALL in your email subject to also convert Data Logs. Note that Data Logs in Excel form may be very large. Remark: The files may also be sent to the server in compressed form (ZIP, TAR). They will be automatically unpacked. Options (such as REPLYEXCEL) will be applied to all files in the email.	X	X
11.	Examine the fault code memory of the EECU for failure. Refer to section 76-10-00.	X	X

The following table has been amended:

05-20-00 Scheduled Maintenance Checks

	GEARBOX	100h	300h
12.	Visually examine the gearbox. Look for leakages at the housing and propeller shaft.	X	X
13.	Measure the gearbox oil level according to section 12-10-00.	X	X
14.	Drain the gearbox oil according to section 85-10-00 para 4.K(a).	X	X
15.	Examine the gearbox oil according to section 85-10-00 para 4.E.	X	X
16.	Collect a sample of used gearbox oil using the oil container provided with the 300 h Service Kit. – Send the oil sample to a suitable lab for analysis (pre-filled out envelope). For further sampling information, refer to SI-E4-013, latest revision.		X
17.	Examine the gearbox oil filter according to section 85-10-00 para 4.G.	X	X
18.	Replace the gearbox oil filter according to section 85-10-00 para 4.G,H. Note: Replace the gearbox oil filter every 900 hours if using Klübersynth GEM 4-68N. If SHELL Spirax S6 GXME 75W-80 was used at any point, the replacement interval remains at 300 h.		X
19.	Refill the gearbox oil according to section 85-10-00 para 4.K(b).	X	
20.	Discard the used oil and fill in new gearbox oil according to section 85-10-00 para 4.K(b). Note: When using Klübersynth GEM 4-68N exchange the gearbox oil every 900h		X

	Activities/Service Interval	100 h	300 h
	BELT DRIVE SYSTEM		
21.	Visually examine the V-Ribbed belt. Look especially for abrasion and signs of damage: – Textile reinforcement visible between the ribs – Tapered ribs – Transverse cracks on the running surface (refer to 85-40-00 1.C for acceptable belt condition) – Dirt between the ribs – Transverse cracks in the ribs – Frayed textile reinforcement. In case of dirt clean the V-Ribbed belt using water only.	X	X
22.	Examine the belt pulley profiles and the tensioner for damage and dirt.	X	X
	ENGINE OIL		
23.	Drain the engine oil according to section 79-20-00 Paragraph 5.A. - Before draining the "old" oil make sure that the LiveView recording described in Step 10 has been performed to collect oil quality data.	X	X
24.	Collect a sample of the used engine oil using the oil container provided with the 100 h Service Kit. – Send the oil sample to a suitable lab for analysis (pre-filled out envelope). For further sampling information, refer to SI-E4-013, latest revision.	X	X
25.	Replace the engine oil filter according to section 79-20-00 Paragraph 3.	X	X
26.	Visually examine the used oil filter for contamination and metal abrasion. – If an abnormal condition is found (metal shavings exceeding 1.5 mm; concentration of particles; clogged filter), this condition must be corrected (e.g. by exchange of defective component within published repair procedures) before further operation of the engine.	X	X
27.	Discard the used oil and fill in new engine oil according to section 79-20-00 para 5.B.	X	X
	FUEL SYSTEM		
28.	Visually examine the fuel system for signs of wear and leakage. Make sure that the fittings are tightened.	X	X
29.	Replace the fuel filter in accordance with the applicable Aircraft Maintenance Manual.	X	X
	COOLING SYSTEM		
30.	Visually examine the cooling system. – Look especially for signs of wear and leakage. – Make sure that the connections are tight fitted, and the hoses are not damaged.	X	X

	Activities/Service Interval	100 h	300 h
	OIL SEPARATOR		
31.	Examine the outlet of the breather line for abnormal quantity of oil (oil dust has to be expected). – In case of oil detection, clean the breather outlet and do an engine ground run according to section 71-00-01. – After the ground run examine the breather outlet again for obvious oil accumulation. – Examine and replace the breather line according to the applicable Aircraft Maintenance Manual.	X	X
	ENGINE		
32.	Disconnect cylinder head drain line from cylinder head below the high-pressure fuel pump. Carefully blow compressed air through the drain hole. Blow compressed air through the cylinder head drain line to make sure that the line is not clogged. Connect the drain line.	X	X
33.	Remove the injector cover according to section 85-10-00 para 3.A.		X
34.	Visually examine the high-pressure fuel lines for signs of wear, leakage and poor condition.		X
35.	Visually examine the Injectors for: - fuel/oil leakage at injectors attachment - fuel leakage at common rail - fuel leakage at fuel return line Do a check of the nuts torque at common rail side. Do a check of the nuts torque at fuel line. Refer to chapter 73 for torque values. NOTE: Any leakage must be documented and reported to Austro Engine GmbH.		X
36.	Examine the oil separator hoses for damage.		X
37.	Examine the oil separator for signs of leakage.		X
38.	Visually examine for coolant loss at the cylinder head.		X
39.	Clean the area below the injector cover using cleaning agent. Make sure that the cylinder head drain is not clogged: The cleaning agent must drain off through the cylinder head drain line. - Do not apply cleaning agent on opened sensor connections.		X
40.	Install the injector cover according to section 85-10-00 para 3.B.		X

	Activities/Service Interval	100 h	300 h
	EXHAUST SYSTEM		
41.	Visually examine the Wastegate Controller for: - Deformations at the connection rod - A deformed mounting eyelet at the rod eye of the waste gate - Mechanical fractures/dents/chafing marks/cracks (affecting the functionality).	x	x
42.	Examine the wastegate controller bridge for: - Loose clamps or cable - Broken or frayed wires - Excessive wear	x	x
43.	Examine the pressure setting of the wastegate controller according to section 81-20-00 para 5.C.	x	x
44.	If MSB-E4-029 has been applied or the latest BPA configuration is installed (refer to section 81-20-00 para 4), clean the wastegate restrictor: - Remove the restrictor. Discard clamps that secure the restrictor. - Use a wire with thickness less than 0.38 mm to remove possible clogging. Therefore, push the wire into the restrictor several times. - Carefully blow with compressed air. - Reinstall the restrictor. Use new clamps. Refer to 81-20-00 para 4.	x	
44.	**Check the tightness of the wastegate actuator membrane for pressure leakage, according to section 81-20-00 para 5.E.** MM-TR-MDC-E4-1021	x	
45.	Examine the exhaust manifold for cracks using a flashlight and a mirror.	x	x
46.	Examine the exhaust manifold gaskets for signs of leakage.	x	x
47.	Check the tightening of the exhaust manifold stud nuts on all 8 positions. Re-tighten the nuts to 30 Nm if necessary.	x	x
48.	Remove the intake air hose from the turbocharger according to the Airplane Maintenance Manual.	x	x
49.	Visually examine the turbocharger for: - Oil leakage - Mechanical wear and damages at the compressor blades - Significant wear and grooves at the air intake - Damage concerning functionality of the turbocharger and sealing (e.g. sealing surfaces…)	x	x
50.	Examine the torque seal indicator (backplate to bearing housing) at the turbocharger. Make sure that the torque seals are not broken. Refer to Fig. 81-4.	x	x
51.	Examine the aircraft oil consumption. The accumulated oil consumption according to the Logs must not exceed 0.1 l/h.	x	x
52.	Install the intake air hose according to the Airplane Maintenance Manual.	x	x

	Activities/Service Interval	100 h	300 h
53.	Renew the torque seal. Refer to Fig. 81-4. – Carefully remove the torque seals. – Clean the surface. The surface at the torque seal positions on the turbocharger must be free from residue. – Do not use degreasing agents! Do not scrape with sharp tools on the turbocharger parts! – Apply the torques seal (Bäder Lacke Schrauben-Sicherungslack, color blue) on the cleaned surface at the turbocharger. The torque seal must cover the compressor and the bearing housing.	X	X
54.	Complete an engine ground run according to section 71-00-01.	X	X
55.	Examine the oil system for leakage. – If leaks are found, examine the root cause and do the repair procedures as published within this maintenance manual. – Measure the engine oil level according to Section 12-10-00.	X	X
56.	Complete the maintenance and record it in the Engine Maintenance Log.	X	X